CSCI 5521: Machine Learning Fundamentals (Spring 2022)[1]

Sample Questions

1. In I-III, select the correct option(s) (it is not necessary to explain).

I. Select all the option(s) that are activation functions:
(a) Sigmoid (b) Gini index (c) Hyperbolic tangent (d) Entropy (e) Lagrangian

II. Select all the option(s) of feature selection strategy to split a decision tree:
(a) Entropy (b) Lagrangian (c) Gini index (d) Misclassification error (e) Kernel

III. Select all the option(s) to control overfitting in decision trees:
(a) Pruning (b) Entropy Threshold (c) Lagrangian (d) Number of leaves (e) Kernel

2. Consider the regularized Multilayer Perceptron (MLP) for binary classification, where each hidden unit uses a sigmoid function. The error function is given by the following expression:

$$E(W, v|X) = -\sum_{t=1}^{N}[r^t \log y^t + (1 - r^t) \log(1 - y^t)] + \frac{\lambda}{2}\sum_{h=1}^{H} ||w_h||_2^2 + \frac{\sigma}{2}||v||_2^2$$

where λ and σ are hyperparameters. Answer the following questions:

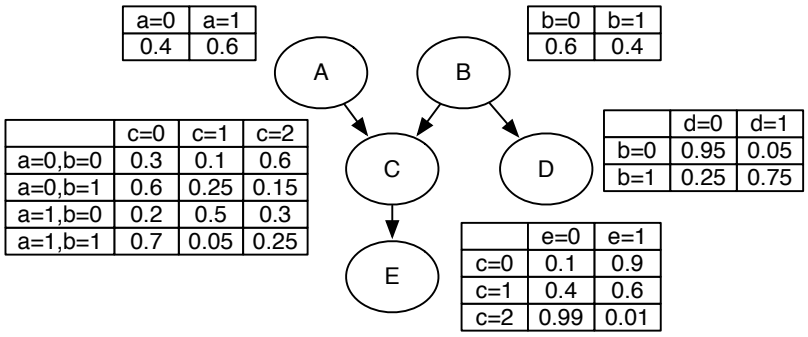
(a) Draw the multilayer perceptron showing: input values $x_0, ..., x_D$, output of hidden units $z_0, ..., z_H$, weights W and v and the output of output unite y.

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(b) Derive the Forward Step equation for y.

(c) Derive the Backward Step equations for w_{hj} and v_h.

3. Build a (Multilayer) Perceptron that calculates a Boolean function (*e.g.*, AND, OR, XOR, NOT, NAND) of its input(s).



4. Consider the above Bayesian Network:

   (a) Find the joint probability $P(A, B, C, D, E)$ as the product of conditional probabilities, according to the graphical model given above.

   (b) Which of the following statements are true? You can select more than one.

   i. $A \perp\!\!\!\perp B|C$

   ii. $C \perp\!\!\!\perp D|B$

   iii. $A \perp\!\!\!\perp B$

   iv. $C \perp\!\!\!\perp D$

   (c) Show how to find the marginal probability $P(A, D)$

(d) Using the conditional probability distribution (CPD) tables in the figure, find:

i. $P(a = 0, b = 1, c = 2, d = 1, e = 1)$

ii. $P(a = 0, d = 1)$

iii. $P(b = 1, c = 2, e = 1 | a = 0, d = 1)$